
Accurate Cardiac Field Reconstruction Can Still Miss Abnormal Tissue

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Abstract

Models that infer cardiac electrical activity from magnetic measurements are often judged by accuracy averaged over the whole tissue, yet scientific usefulness depends on locating abnormal tissue. We test whether global accuracy reflects recovery of abnormal tissue in a controlled benchmark with 500 independent cardiac tissue simulations. Fourier neural operator (FNO) and DeepONet models reconstruct local activation time (LAT), action potential duration (APD), and a mask of abnormal tissue from synthetic magnetic recordings. On 75 test cases, the FNO and DeepONet models selected for field accuracy achieve LAT and APD90 $R^2 \geq 0.954$; their Dice overlaps are 0.578 and 0.545. In 18 cases with slowed conduction, the same fixed weights retain global field R^2 between 0.978 and 0.986 while Dice falls to 0.167 and 0.163. A separate staged diagnostic detects conduction information in the magnetic measurements and in predicted LAT from a refinement model fixed using validation data; failure appears during final region extraction. Together, these results identify localization as the main failure point despite accurate global field reconstruction. Global accuracy can substantially overstate biological usefulness.

1 Introduction

Magnetocardiography records the weak magnetic field generated by cardiac electrical activity [1–4]. Inferring the underlying electrical pattern is difficult because several internal patterns can explain similar external measurements [5]. Related electrocardiographic imaging studies have shown the value of noninvasive activation maps in both simulation and patients [6–9]. Neural operators learn a transformation from an observed field to a target field [10]. FNOs learn this transformation in Fourier space [11]; DeepONet combines a representation of the input measurement with the coordinates at which the output is requested [12]. Training with physical constraints can also connect neural models to governing equations [13, 14]. Our models use supervised losses throughout.

A high global field score can coexist with failure to recover a small pathological region. Most pixels describe normal background tissue, so a smooth prediction can score well while missing the location, center, or physiological contrast of an abnormal region. Biomedical image analysis has repeatedly shown that a single aggregate metric can hide errors that matter to the task, particularly for small structures and imbalanced regions [15–18]. This concern is especially important in biological machine learning, where evaluation and reporting choices can make results appear more reliable than they are [19, 20].

We study this mismatch in a controlled synthetic cardiac magnetic benchmark. First, we compare FNO and DeepONet using global field accuracy and overlap with abnormal tissue. Second, we show that nearly perfect global fields can coexist with poor localization of conduction slowing. Third, we trace the error through the pipeline to determine where useful information is lost. Augmented versions of one simulation are kept in the same data partition to prevent leakage [21].

38 **Anonymous code and results.** The source code, evaluation scripts, documentation, and integrity
39 manifests are available in the [anonymous code repository](#). Test arrays, selected checkpoints,
40 bootstrap outputs, and provenance records are in the [anonymous OSF results archive](#).

41 2 Controlled cardiac magnetic benchmark

42 **Synthetic electrophysiology and measurements.** From a generated 5,000-case upstream corpus,
43 we selected a deterministic, condition-stratified pilot of 500 independent successful simulations:
44 75 homogeneous, 125 with altered repolarization, 125 with slowed conduction, and 175 with both
45 changes. Each case uses a fixed 2D ventricular sheet measuring 10 mm by 10 mm, the Bueno-
46 Orovio–Cherry–Fenton minimal human ventricular model [22], and the openCARP monodomain
47 environment [23]. This setup follows established cardiac tissue modeling practice [24–27]. Each
48 simulation yields LAT, APD80, APD90, and the true abnormal-tissue mask.

49 For each time point and triangular element, we approximate the tissue current as $J_{eq} = -D_{eff}\nabla V_m$
50 and use the Biot–Savart relation to calculate the external magnetic field. The resulting signal in
51 relative units represents the external magnetic field produced by equivalent current in the sheet. The
52 primary input combines a magnetic sequence measured 50 mm above the sheet with an axial gradient
53 sequence computed by finite differences over a 30 mm baseline centered on the same plane. Both use
54 a 32×32 sensor grid over 301 time points. Validity masks exclude missing sensor values during
55 normalization and set them to zero before inference. Other measurement settings vary noise and
56 masking while preserving the underlying biological case.

57 **Models.** The FNO maps the input to 64 features and applies four 2D Fourier blocks with 12 modes
58 per spatial axis (4.82 million trainable parameters). DeepONet uses a convolutional branch and a
59 trunk with Fourier coordinate features, 64 basis functions, and 96-dimensional representations (0.287
60 million trainable parameters). Both predict LAT, APD80, APD90, and abnormal-tissue probability.
61 We use APD90 as the headline repolarization endpoint to limit duplicate reporting; APD80 is a
62 secondary consistency check. We trained each configuration once with seed 20260525 using input
63 normalization within each sample, AdamW, batch size 6, dropout 0.08, learning rate 8×10^{-4} ,
64 weight decay 10^{-4} , at most 10 epochs, and early stopping patience 3. DeepONet samples 256 spatial
65 queries per training example. Some localization variants also use summary statistics derived from the
66 magnetic measurements. At inference, both models receive masked magnetic measurements; FNO
67 also receives grid channels for spatial coordinates, while DeepONet receives output query coordinates
68 through its trunk. Training uses supervised objectives throughout.

69 **Split, selection, and statistics.** All measurement variants from one simulation remain in one
70 partition: 350 training, 75 validation, and 75 test cases, corresponding to 1,050/225/225 input rows.
71 The test set contains 12 homogeneous, 18 repolarization-only, 18 conduction-only, and 27 mixed
72 cases. The final analysis uses one predeclared clean input with two active measurement channels per
73 test simulation; two inactive slots are zeroed to retain a four-slot tensor layout. For each architecture,
74 the saved model chosen for field accuracy maximizes mean validation LAT/APD90 R^2 , while a
75 separate model chosen for localization maximizes validation Dice at a fixed 0.5 threshold. Both saved
76 models predict fields and the probability of abnormal tissue. Our primary paired analysis computes
77 every endpoint from the fixed weights chosen for field accuracy. Models chosen for localization
78 appear only in the secondary analysis of selection effects. We freeze every saved model and threshold
79 before evaluating the test set.

80 Global LAT and APD90 R^2 pool valid pixels across the 75 cases. Localization endpoints are pooled
81 Dice over the 63 abnormal cases and APD90 contrast inside versus outside the predicted region.
82 The latter is a generic region-output diagnostic; the conduction analysis uses LAT within true or
83 target LAT-anomaly support because slowed conduction primarily changes activation timing and
84 LAT gradients. Center error and false abnormal area in homogeneous cases are secondary release
85 diagnostics. An empty or full predicted region receives zero recovered contrast and is also counted
86 explicitly. We report 95% percentile intervals from 5,000 paired bootstrap samples of independent
87 simulations [28]; they quantify uncertainty from sampling the test cases. Bootstrap resampling draws
88 simulation identifiers, keeping all augmented rows from a case together.

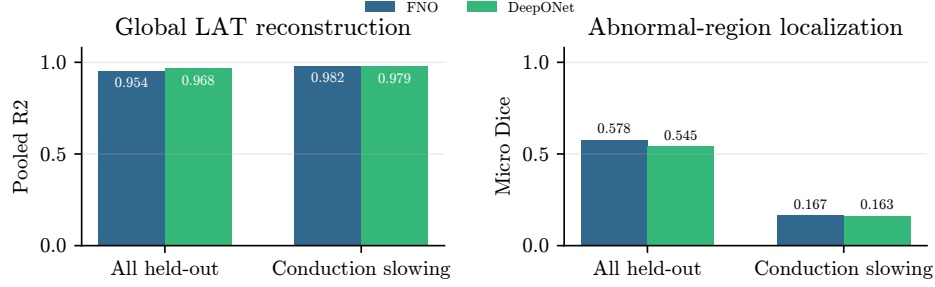


Figure 1: Global LAT accuracy and abnormal tissue overlap from the same saved model. The full cohort has $n = 75$ for LAT and $n = 63$ for Dice; conduction has $n = 18$. Exact 95% bootstrap intervals are in the text.

89 3 Global accuracy conceals localization failure

90 Across 75 test cases, the FNO and DeepONet models chosen for field accuracy reach LAT R^2 of
 91 0.954 [0.941, 0.965] and 0.968 [0.960, 0.975]. APD90 R^2 is similarly high at 0.960 [0.944, 0.971]
 92 and 0.970 [0.966, 0.974]. These results would make either fixed model look reliable if evaluation
 93 stopped at a global field score. The same weights tell a different localization story: pooled Dice is
 94 0.578 [0.498, 0.648] for FNO and 0.545 [0.466, 0.614] for DeepONet (Figure 1). Mean per-case
 95 Dice is 0.470 and 0.452, with zero overlap in 12 and 14 of 63 abnormal cases. The secondary APD80
 96 check gives global R^2 of 0.957 and 0.976. The R^2 for APD90 contrast inside versus outside each
 97 model’s predicted region is -0.109 and 0.095 , with intervals that include zero. The combination of
 98 high field scores, weak overlap, and unstable contrast exposes a clear gap between reconstruction and
 99 localization.

100 The disagreement is most pronounced for conduction slowing. On 18 independent conduction cases,
 101 the same FNO weights chosen for field accuracy produce LAT/APD90 R^2 of 0.982/0.986 and Dice
 102 of 0.167 [0.057, 0.270]. The same DeepONet weights produce 0.979/0.978 and Dice of 0.163 [0.070,
 103 0.257] (Figure 1). Their APD90 contrast R^2 values are -0.286 [$-3.230, 0.397$] and -0.376 [$-2.834,$
 104 0.504], respectively. Both intervals cross zero and extend far into negative values, indicating unstable
 105 contrast recovery. Each model also produces an empty or full region in 5 of 18 cases. Figure 2 makes
 106 the mismatch visible for one disclosed example chosen after analysis: both LAT maps score above
 107 0.98 while both thresholded regions have zero overlap with the true region. Selection by localization
 108 performance on validation data produces conduction Dice of 0.006 for FNO and 0.333 for DeepONet.
 109 Because these values come from different checkpoints, we report them separately from the R^2 values
 110 obtained with the models chosen for field accuracy.

111 4 Where the conduction result fails

112 We next trace information through three stages without additional training, using a separate DeepONet
 113 refinement checkpoint fixed using validation data. First, a 500-tree random forest [29], chosen
 114 by validation balanced accuracy, uses 50 magnetic descriptors only, a minimum leaf size of two,
 115 balanced-subsample class weights, and one clean row per simulation. It detects conduction slowing
 116 with test balanced accuracy 0.822 [0.727, 0.907] across 75 cases, showing that the measurements
 117 contain information about the condition. Second, on the 18 conduction cases, the refinement model’s
 118 predicted LAT recovers the quantity associated with conduction inside the true abnormal region, with
 119 $R^2 = 0.954$ [0.919, 0.974]. Using support derived from target LAT anomalies gives $R^2 = 0.948$
 120 [0.615, 0.966]. Calculations with these support masks serve as internal diagnostics for tracking where
 121 information disappears along the pipeline.

122 The final stage collapses. When its predicted region is used, the recovered quantity has $R^2 \approx 0$,
 123 and every conduction case produces an empty region. Figure 3 summarizes the progression from
 124 informative magnetic summaries to accurate local LAT recovery and then unusable region output.
 125 The generic region extractor is the observed collapse point for this checkpoint.

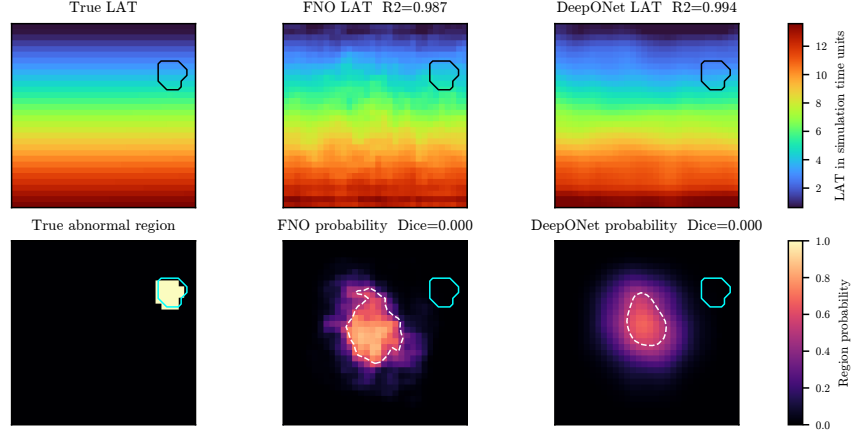


Figure 2: Illustration chosen after analysis from the same weights used for field accuracy. True regions are outlined in black or cyan and predicted regions in dashed white. After completing model selection, we chose the conduction case with the largest joint discrepancy between high LAT accuracy and low Dice across both models.

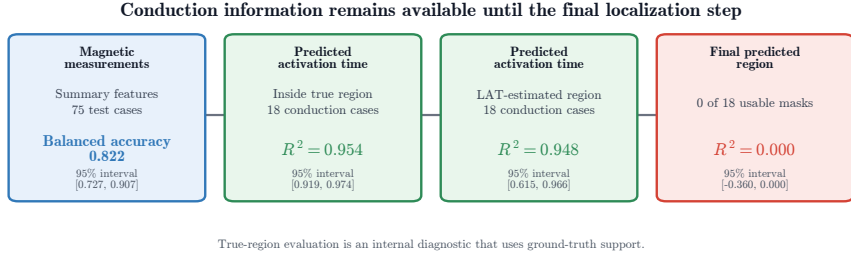


Figure 3: Diagnostic across three stages using a refinement model fixed with validation data. Magnetic summaries distinguish conduction cases, and predicted LAT retains the target quantity when evaluated inside the true region. The final stage, which uses the predicted region, collapses in all 18 cases.

Interpretation. Global R^2 weights every valid pixel, allowing background tissue to dominate a small abnormal region. A shared region output must represent repolarization changes in APD and conduction changes in LAT gradients, so smooth fields can coexist with erased local patterns. Same-model results, degenerate masks, and unstable contrast establish this mismatch; the refinement checkpoint collapses at region extraction.

5 Limitations and practical takeaway

The benchmark uses a fixed 2D sheet, synthetic fields in relative units, planar sensors, and simulated noise and missingness. It lacks anatomy from individual patients, clinical or real SQUID data, and training constrained by physical equations. All neural results come from one seed. FNO stopped after eight epochs, while DeepONet reached the 10-epoch cap with validation loss and sampled Dice still improving. Retraining variability and possible localization undertraining therefore remain unresolved. The 18 conduction cases give wide contrast intervals. Every primary endpoint uses one model per architecture chosen for field accuracy; separately chosen models are labeled secondary. The refinement diagnostic covers one checkpoint and cannot establish a universal mechanism across architectures.

Next tests should separate conduction and repolarization outputs, emphasize boundaries and centers, and select thresholds on validation data. Global field R^2 alone is unsafe for biological localization; benchmarks should also report overlap, center error, local physiological contrast, and condition-specific results [18, 20].

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Large language model use disclosure

Large language models were used for manuscript refining, figure generation, and as code assistants. Scripts checked the claims against the saved result files and file integrity records used for the final analysis. The authors remain responsible for the core manuscript writing, scientific claims, analysis, accuracy, and integrity of the submission.

234 **Reproducibility statement**

235 The anonymous code and result archives contain grouped splits, selection records based only on
236 validation data, selected checkpoints, training logs, prediction maps for the test cases, bootstrap
237 outputs, dependency snapshots, licenses, and file checksums. A compact command recomputes
238 every primary point estimate from the saved maps and checks it against the authoritative table for
239 the same saved models. The remaining 4,500 upstream cases are omitted because of their size; the
240 released pilot records and test arrays are sufficient to verify the paper’s primary results. The reported
241 confidence intervals cover sampling among the test cases, and the release contains one training seed
242 per model.

243 **Experimental details and compute**

244 Primary FNO and DeepONet runs used the settings stated in Section 2. The DeepONet loss weights
245 were 1.0 for fields, 1.0 for the region map, 0.45 for presence, 0.70 for region parameters, 0.35 for
246 area, 0.08 for condition, 0.02 for consistency, and 0.04 for scalar outputs. Training and evaluation
247 used one NVIDIA RTX 5090 with 32,607 MiB memory on an AMD Ryzen 9 9950X3D host with 16
248 cores and 61.6 GiB RAM, PyTorch 2.11.0, and CUDA 12.8. The complete data pipeline for 5,000
249 cases used eight CPU workers and took 6.13 hours. The historical logs do not contain exact elapsed
250 time or energy use for each neural training run.

251 **Broader impact and ethics**

252 Evaluation tied to the biological task could reduce the risk that globally accurate cardiac maps
253 are mistaken for reliable localization. Conversely, premature clinical use of a synthetic model
254 could misidentify abnormal tissue and influence care. We mitigate this risk through explicit failure
255 documentation and a scope limited to research. The study consists entirely of simulations and includes
256 no participants or personal data.

257 **Asset licenses**

258 The release documents software licenses. External datasets and openCARP binaries remain outside
259 the release. openCARP is used under Academic Public License 1.1 for noncommercial academic
260 research; carputils uses Apache 2.0 and meshalyzer uses GPLv3. New code is released under MIT
261 and generated result artifacts under CC BY 4.0.

262 **A Sensitivity to sensor distance**

263 We evaluated the trained models without retraining after recomputing synthetic measurements at
264 distances of 35, 50, and 65 mm from the tissue. Relative to the nominal 50 mm plane, the 65
265 mm FNO LAT/APD90 R^2 changes were $-0.0737/ - 0.1082$, whereas DeepONet changed by
266 $+0.0002/ + 0.0002$ (Figure 4). Normalization within each sample and the fixed planar sheet limit
267 this result to synthetic measurement sensitivity; generalization to patient geometry remains untested.

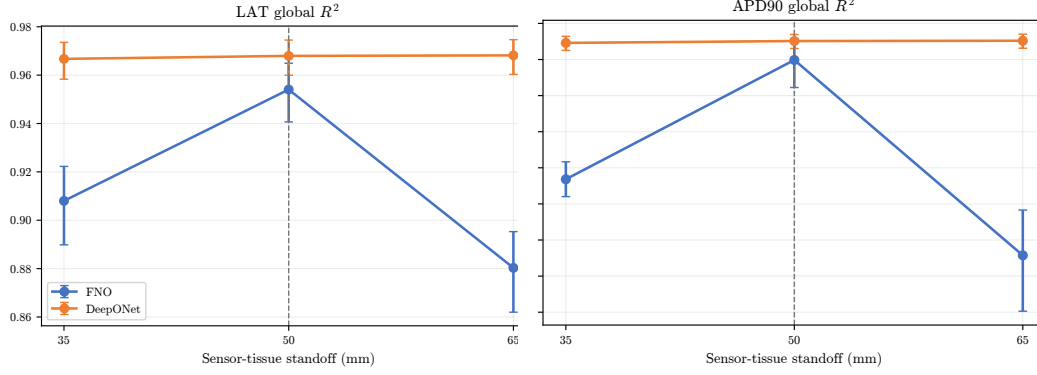


Figure 4: Sensitivity of global field reconstruction to the distance from sensor to tissue without retraining. The dashed line marks the nominal 50 mm plane; error bars are 95% bootstrap intervals over independent simulations.

B Evaluation metric definitions

Global R^2 is recomputed within each bootstrap sample after pooling valid pixels. Pooled Dice aggregates pixels across all abnormal cases; mean per-case Dice averages the overlap computed separately for each case. Center error is Euclidean distance in normalized sheet coordinates. APD90 regional contrast is the mean predicted APD90 inside minus outside the predicted region. An empty or full region receives zero recovered contrast so localization failures remain in the paired analysis. All confidence intervals sample independent simulation identifiers with replacement while preserving model, condition, target, and measurement pairing.

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

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